

DSC 335 Exam 2 QMD Part

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Question 1 (13 points)

The *births14* data set in the **openintro** R package gives information on a random sample of human births recorded by the US Department of Health and Human Services in 2014. Our response variable is the *weight* of the babies, in pounds. For this problem we will consider the following predictors: the *sex* of the baby (male or female, with female as the baseline level), the mother's age (*mage*), and the number of *weeks* the mother was pregnant before giving birth.

a. Consider the following prior information: For children of the same sex and mother's age, we think that for each additional week of pregnancy the weight of the baby will increase by 0.5 pounds, but that the increase could plausibly be anywhere from 0.1 to 0.9 pounds. We also think that male babies will weigh around 0.4 pounds more than female babies born at the same number of weeks to mothers of the same age, and that this difference could plausibly be between 0.2 and 0.6 pounds. Finally, we think that a $N(0, 1)$ distribution is a reasonable prior for the coefficient of the *mage* variable. Using this prior information and putting default priors on all other parameters, fit a Normal Bayesian regression model in the code chunk below. (If you are unable to figure out how to incorporate the priors, you can get partial credit for part (a) but potentially full credit for the rest of the question if you run the code with default priors.)

```
model <- stan_glm(weight ~ sex + mage + weeks, data = births14,
                  family = gaussian,
                  prior = normal(location = c(.4, 0, .5), scale = c(.1, 1, .2)), chains = 4,
```

b. Based on 90% posterior credible intervals, which of the variables are significant predictors of a baby's weight, given the other variables in the model. Justify your answer.

ANS: All the variables are significant predictors because zero is not included in any of the 90% posterior credible intervals.

```
posterior_interval(model, prob = .9)
```

	5%	95%
(Intercept)	-5.268101792	-3.48847488
sexmale	0.297564841	0.48320426
mage	0.007058564	0.02671092
weeks	0.260141616	0.30371214
sigma	1.035358457	1.11498902

c. Calculate a 95% posterior predictive interval for the weight of a male baby born to a 30 year old mother whose pregnancy lasted 37 weeks.

```
pred <- posterior_predict(model, newdata = data.frame(sex = 'male', mage = 30, weeks = 37))
posterior_interval(pred, prob = .95)
```

	2.5%	97.5%
1	4.880132	9.089762

Question 2 (10 points)

The *q2data* data set created in the code chunk below contains the variable *median_edu* which gives the median education level of counties in the US. It has four levels: *below_hs*, *hs_diploma*, *some_college*, and *bachelors*. It also contains the following predictors: the percent of the population below the *poverty* line, the population of the county in 2017 (*pop2017*), and *metro*, which is yes if the county has a metro area and no if it does not. We will use the *q2data* to classify the median education level of Boyle County, which had a 2017 population of 29924, a poverty level of 15.5, and does not have a metro area.

a. Fit a Naive Bayes model that only includes the *poverty* and *pop2017* variables. How would this model classify the median education level of Boyle County?

ANS: With the assumptions that the predictors are independent, it calculates a conditional probability of each target class given the predictor variables with a simple application of Bayes rule. These probabilities aren't posterior results but a relative belief of one class over another.

means which category does it assign to Boyle

```
model <- naiveBayes(median_edu ~ poverty + pop2017, data = q2data)
```

b. Fit a Naive Bayes model that includes the *poverty*, *pop2017*, and *metro* variables. How would this model classify the median education level of Boyle County?

ANS: With two quantitative and one qualitative predictor, the estimator assumes that these quantitative variables are normally distributed with some mean and variance and that they are independent of each other. Naive bayes produces conditional probabilities for each target class given the predictors.

```
model2 <- naiveBayes(median_edu ~ poverty + pop2017 + metro, data = q2data)
```

c. Now consider the in-sample (not cross-validated) performance of the two models above. Which did a better job of classifying the counties whose median education level is some_college? Justify your answer.

ANS: Although both models did a poor job in predicting target labels for all 4 classes, the second model has a slightly higher accuracy than the first. The sensitivity of target label 'some_college' is equal to 22.8% in first model and 38% for the second. Hence, the second model does a better job in classifying counties whose median education level is some_college.



```
naive_classification_summary(model, data = q2data, y = 'median_edu')
```

```
$confusion_matrix
  median_edu  below_hs      hs_diploma some_college bachelors
  below_hs 0.00% (0) 100.00%      (2) 0.00% (0) 0.00 (0)
  hs_diploma 0.21% (3) 98.00% (1,369) 1.79% (25) 0.00
  some_college 0.65% (11) 76.58% (1,295) 22.77% (385) 0.00
  bachelors 2.17% (1) 36.96% (17) 60.87% (28) 0.00

$overall_accuracy
[1] 0.5593112
```

```
naive_classification_summary(model2, data = q2data, y = 'median_edu')
```

```
$confusion_matrix
  median_edu  below_hs      hs_diploma some_college bachelors
  below_hs 0.00% (0) 100.00%      (2) 0.00% (0) 0.00 (0)
  hs_diploma 0.36% (5) 94.42% (1,319) 5.23% (73) 0.00
  some_college 0.47% (8) 61.38% (1,038) 37.97% (642) 0.18 (3)
  bachelors 0.00% (0) 26.09% (12) 69.57% (32) 4.35 (2)

$overall_accuracy
[1] 0.6259566
```

Question 3 (17 points)

Dr. Daryl is concerned that students aren't attending his office hours enough so one semester he decides to track the number of students that come to his office during each 1 hour period. Prior to the semester starting, he thinks that the average number of students that come each hour could be modeled by a $\text{Gamma}(1, 0.5)$ distribution. He records the first 10 office hour attendances as follows: 1,3,0,4,0,2,2,0,3,1.

a. In the code chunk below, fill in the missing information (wherever there is a `?`) to perform an MCMC simulation of his posterior understanding of the true average number of students λ that show up to a single office hour, based on a Gamma-Poisson model. Note: If you cannot get your simulation to run, you can determine the posterior distribution analytically and use that for the rest of this problem.

```
# STEP 1: DEFINE the model
gp_model <- "
  data {
    int<lower = 0> Y[10];
  }
  parameters {
    real<lower = 0> lambda;
  }
  model {
    Y ~ poisson(lambda);
    lambda ~ gamma(1, .5);
  }
"

# STEP 2: SIMULATE the posterior
gp_sim <- stan(model_code = gp_model, data = list(Y = c(1,3,0,4,0,2,2,0,3,1)),
              chains = 4, iter = 5000*2, seed = 84735)
```

b. Dr. Daryl wants to use his posterior distribution to test the hypotheses $H_0 : \lambda \geq 2$ versus $H_a : \lambda < 2$. Calculate the posterior probability of both hypotheses.

```
mean(extract(gp_sim)$lambda >= 2)
```

```
[1] 0.1623
```

```
mean(extract(gp_sim)$lambda < 2)
```

```
[1] 0.8377
```

c. Run the code chunk below to obtain the prior probabilities of the hypotheses. Use these probabilities and your answers from part b to calculate the Bayes Factor for the alternative hypothesis.

```
pgamma(2,1,0.5)
```

```
[1] 0.6321206
```

```
1-pgamma(2,1,0.5)
```

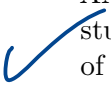
```
[1] 0.3678794
```



```
((mean(extract(gp_sim)$lambda < 2)) / (mean(extract(gp_sim)$lambda >= 2))) / (pgamma(2,1,0.5)
```

```
[1] 3.003832
```

d. Summarize the strength of the evidence for the alternative hypothesis based on your answers to parts b and c.



ANS: From the posterior distribution, there's an 83% probability that true average number of students that show up to a single office hour is less than 2. With a calculated bayes factor of 2.89, there's slight evidence to suggest an increase in the plausibility of the alternative hypothesis in light of the observed data.



e. Suppose our hypotheses had been $H_0 : \lambda = 2$ versus $H_a : \lambda \neq 2$ instead. Describe a method for conducting such a test (but do not actually perform the test).

ANS: Two-sided hypothesis test.

f. (Bonus) The MCMC procedure above produces 20,000 λ values, in four chains of 5,000. Find the first and second values of λ produced by the first chain. Calculate the likelihood of both of those values, given the data. Use these likelihoods to describe why the Metropolis Hastings algorithm moved from the first value to the second value.